

Database Project – Online Coding Contest Platform

Objective

The objective of this project is to design and implement a database-driven **Online Coding Contest Platform** where users can participate in coding contests, solve programming problems, practice algorithms, improve their coding skills, and compete with others through rankings and ratings. The platform provides contests, practice mode, editorials, badges, notifications, and a rating system to enhance the competitive programming experience.

Application Users

1. **Unregistered Users (Visitors)** – Can explore the platform without signing in.
2. **Users (Students/Coders)** – Registered users who can participate in contests, solve problems, earn ratings, and receive badges.
3. **Problem Setters** – Create contests, add problems, test cases, editorials, and manage contest content.
4. **Admin** – Platform administrators who manage users, contests, problems, and overall platform functionality.

Use Cases – Unregistered Users (Visitors)

Users who visit the platform without logging in can:

1. Explore Programming Problems
 - View problem titles and difficulty levels.
 - Browse available practice problems.
2. View Contest Schedule
 - See upcoming and past contests.
 - View contest duration and start time.
3. Browse Leaderboard
 - View top-rated users.
 - Explore user rankings.
4. View Editorial Availability
 - Check whether editorials have been released.

Use Cases – Registered Users (Coders)

Registered users can:

1. Register & Manage Profile
 - Name
 - Email
 - College
 - Rating
 - Password
 - Profile information
2. Register for Coding Contests.
3. Participate in Live Contests.
4. Solve Problems in Practice Mode.
5. Submit Solutions using different programming languages.
6. View Submission History and Results.
7. Participate in Virtual Contests.
8. Track Rating History after every contest.
9. Earn Badges and Achievements.
10. Receive Notifications for contests, editorials, and announcements.
11. Read Editorials after release.

User Registration

To register, users provide:

- User ID (Unique)
- Name
- Email
- Password
- College
- Role
- Rating
- Account Creation Date
- Etc.

Use Cases – Problem Setters

Problem setters can:

1. Create and manage coding contests.
2. Create programming problems.

3. Add test cases for each problem.
4. Assign tags and difficulty levels.
5. Publish editorials after contests.
6. Monitor submissions and plagiarism reports.
7. Manage contest announcements.

Problem Setter Registration

To register, problem setters provide:

- Setter ID (Unique)
- Name
- Email
- Password
- Organization
- Experience (Optional)
- Account Status
- Contact Information
- Etc.

Use Cases – Admin

Admins oversee the entire platform and can:

1. Manage Users and Problem Setters.
2. Activate or Deactivate user accounts.
3. Create and manage programming languages.
4. Monitor contests and submissions.
5. Manage badges and achievements.
6. View reports and platform statistics.
7. Manage notifications.
8. Monitor ratings and contest leaderboards.

Key Features & Functionalities

1. Contest Management

- Create live coding contests.
- Register users for contests.
- Maintain contest leaderboards.

2. Problem Management

- Add programming problems.
- Assign difficulty and tags.
- Add multiple test cases.

3. Submission System

- Submit code in multiple programming languages.
- Automatic evaluation.
- Verdict generation.
- Runtime and memory tracking.

4. Rating System

- Rating updates after every contest.
- Rating history maintenance.
- Global rankings.

5. Practice Mode

- Solve problems without participating in contests.
- Maintain separate practice submissions.

6. Editorial System

- Release editorials after contests.
- Explain optimal solutions.

7. Gamification

- Earn badges.
- Rating progression.
- Contest rankings.
- User achievements.

8. Notification System

- Contest reminders.
- Editorial release notifications.
- Badge earned notifications.
- Platform announcements.

Reports & Queries

♦ For Users

1. View submission history with verdicts.
2. View rating history.
3. Find unsolved problems in registered contests.
4. View earned badges.
5. View contest rankings.

♦ For Problem Setters

6. Find the most attempted problems.
7. View acceptance rate of each problem.
8. View submissions for a contest.
9. View problems with their tags.
10. View contest participation statistics.

♦ For Admins

11. Find the top-rated users.
12. Find contests with the highest participation.
13. Find users who have never participated in a contest.
14. View total submissions by programming language.
15. Find the most popular problem tags.
16. Generate platform usage reports.
17. View notification statistics.

Conclusion

This **Online Coding Contest Platform** provides a complete environment for competitive programming. By integrating contest management, problem solving, submissions, leaderboards, ratings, badges, notifications, and editorials, it enables users to improve their programming skills while offering administrators and problem setters efficient tools to manage contests and platform activities. The system ensures a scalable, engaging, and organized competitive programming experience similar to platforms such as Codeforces and LeetCode.